

Introduction to STM32 GPIO

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Unless stated otherwise, for the second part of the module,

‘STM32 Chip’ = STM32L4S5VIT6

‘Board’ = B-L4S5I-IOT01A

Acknowledgement: Slides adapted from Dr Rajesh Panicker



OVERVIEW

- What is GPIO?
- GPIO Organization
- GPIO Registers

GPIO ORGANIZATION IN STM32L4Sxxx

- GPIO: General Purpose Input-Output
- STM32L4Sxxx have up to 140 I/O lines, organized into ports and pins
- GPIO ports:
 - GPIOA to GPIOI (up to 16 I/O pins per port)
 - Exact number of pins implemented depends on the specific SoC (System-on-chip)

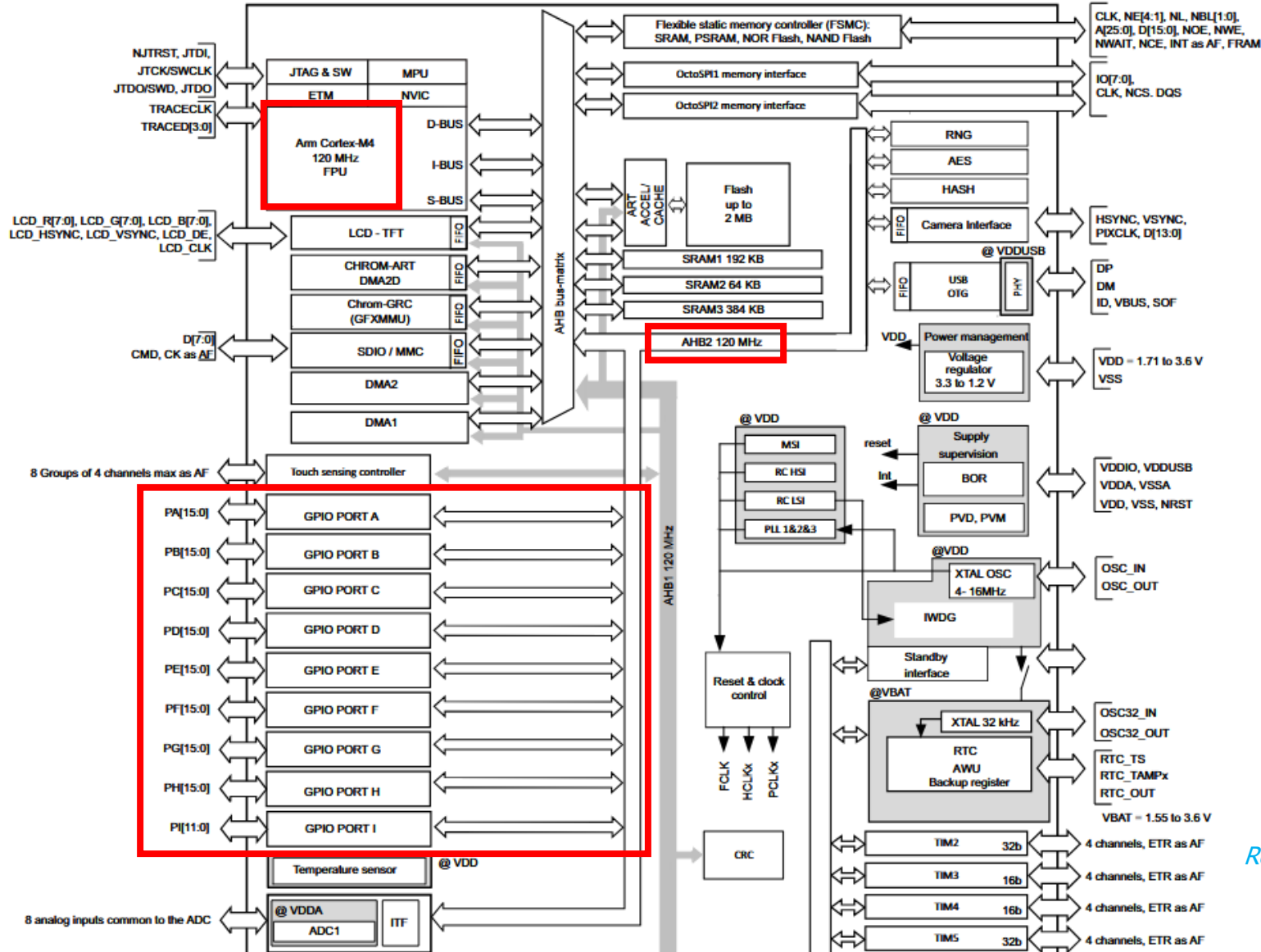
Table 2. STM32L4S5xx, STM32L4S7xx and STM32L4S9xx features and peripheral counts (continued)										
Peripheral	S5VI	S7VI	S9VI	S5QI	S5ZI	S7ZI	S9ZI	S5AI	S7AI	S9AI
GPIOs	83		77	110	115		112 ⁽³⁾	140		131

- For STM32L45SVI:
 - 83 GPIOs
 - Port A to Port E -> 16 pins each
 - Port H -> 3 pins

GPIO ORGANIZATION IN STM32L4Sxxx MCU

- Direct connection to AHB2 bus (120 MHz) allows fast I/O timing
- GPIO peripheral is used to interface with
 - Inputs like Switches
 - Outputs like LEDs
 - Embedded/on-chip peripherals for both digital & analog signals

GPIO BLOCK DIAGRAM (PARTIAL) - STM32L4Sxxx

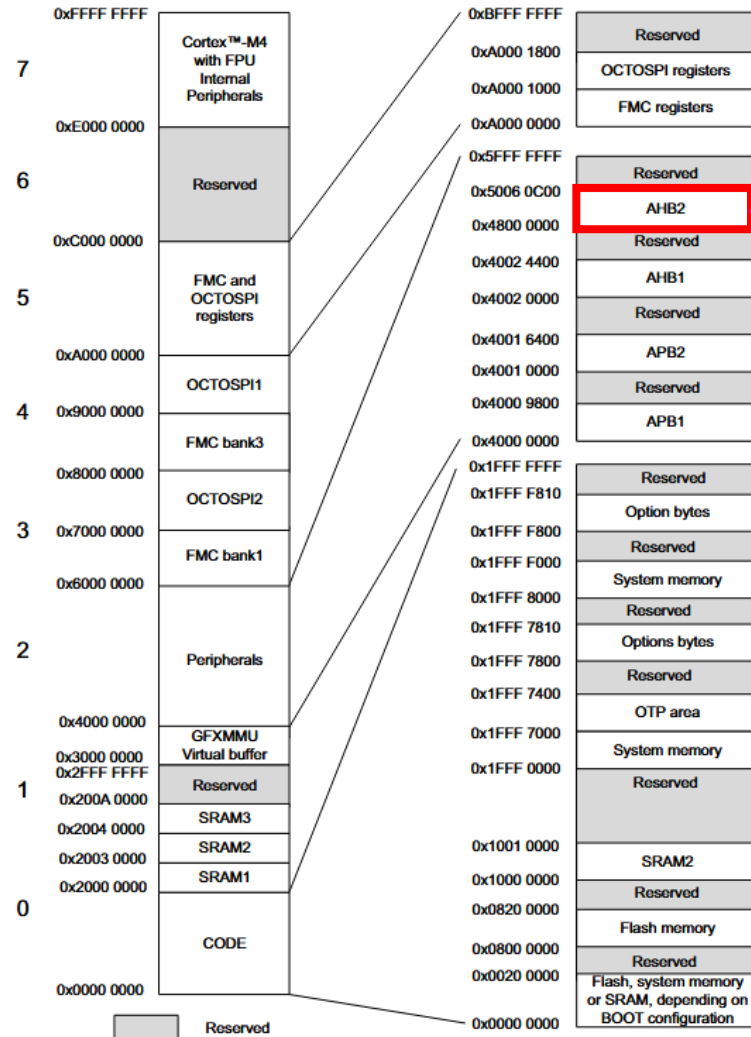


Refer to DS12024: STM32L4S5VI datasheet, Fig 1 (pg. 16)

GPIO MULTIPLEXING IN STM32

- Each pin of the GPIO ports can be individually configured by software in several modes & directions:
 - Input
 - Output
 - Analog
 - Alternative function (AF)
- The number of physical pins are kept low by multiplexing different functions on a particular pin – helps to keep the cost low
- Possible mutual exclusivity of functions, but typical applications need only a small subset of the various peripheral interfaces available on an SoC
- GPIO lines are configurable through GPIO registers, which have to be accessed as 32-bit words, half-words or bytes

GPIO REGISTERS - STM32L4S+++



Bus	Boundary address	Size (bytes)	Peripheral
AHB2	0x4800 2000 - 0x4800 23FF	1 KB	GPIOI
	0x4800 1C00 - 0x4800 1FFF	1 KB	GPIOH
	0x4800 1800 - 0x4800 1BFF	1 KB	GPIOG
	0x4800 1400 - 0x4800 17FF	1 KB	GPIOF
	0x4800 1000 - 0x4800 13FF	1 KB	GPIOE
	0x4800 0C00 - 0x4800 0FFF	1 KB	GPIOD
	0x4800 0800 - 0x4800 0BFF	1 KB	GPIOC
	0x4800 0400 - 0x4800 07FF	1 KB	GPIOB
	0x4800 0000 - 0x4800 03FF	1 KB	GPIOA

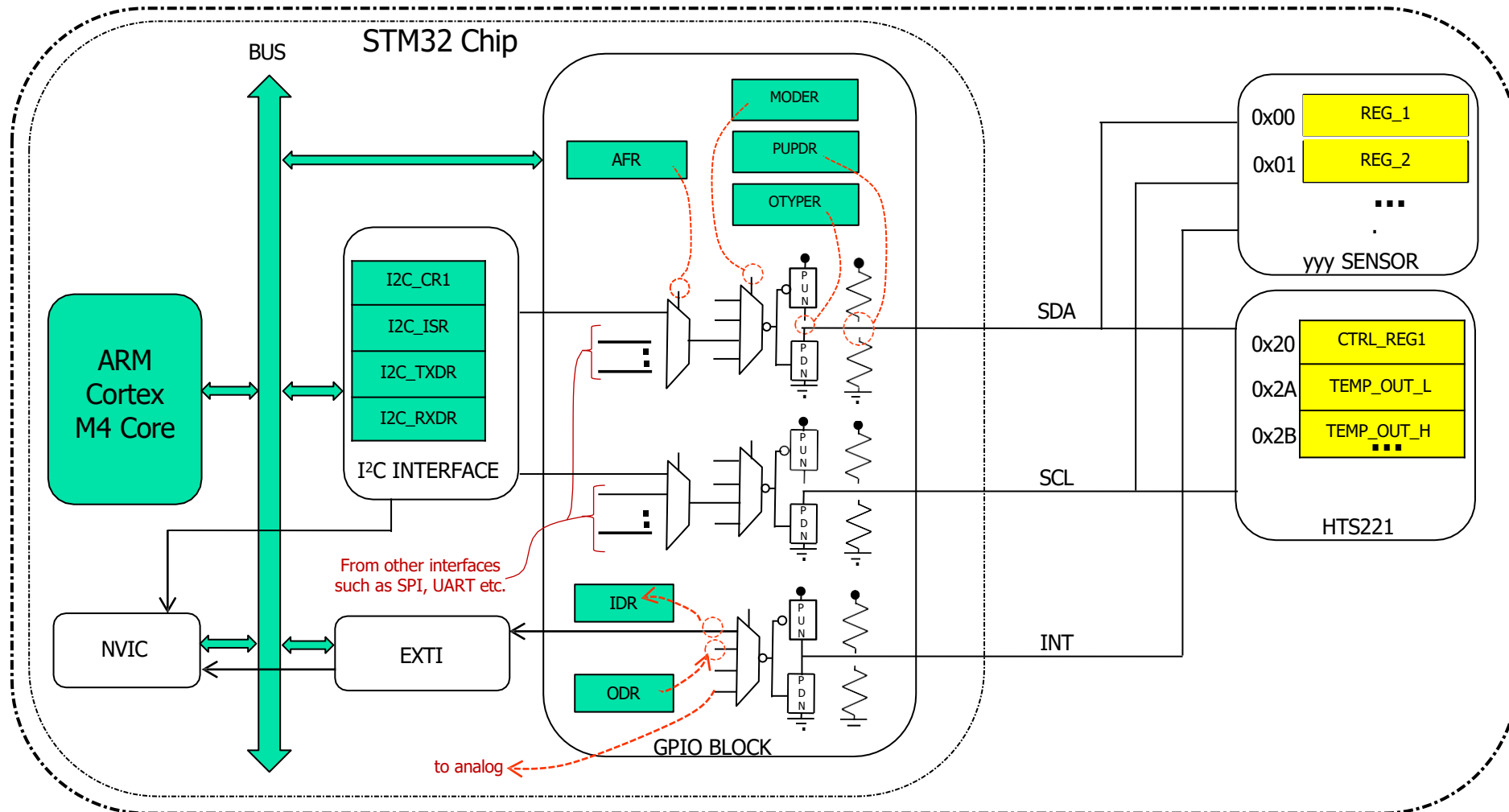
STM32L4Sxxx devices peripheral register boundary addresses

Memory map for STM32L4Sxxx devices

GPIO REGISTERS OVERVIEW - STM32L4S5VI

Type	Register Name (where x = A to I)			Offset	Description
Control	Mode register	GPIO x _MODER	RW	0x00	Selects pin mode: Input, Output, Analog, or Alternate Function
	Output type register	GPIO x _OTYPER	RW	0x04	Configures output type: Push-Pull or Open-Drain
	Output speed register	GPIO x _OSPEEDR	RW	0x08	Sets output speed: Low, Medium, High, or Very High
	Pull-up / Pull-down register	GPIO x _PUPDR	RW	0x0C	Enables internal Pull-Up or Pull-Down resistors
	Configuration lock register	GPIO x _LCKR	RW	0x1C	Locks pin configuration (prevents accidental changes)
	Alternate function low register	GPIO x _AFRL	RW	0x20	Selects Alternate Function (AF0 to AF15) for pin 0-7
	Alternate function high register	GPIO x _AFRH	RW	0x24	Selects Alternate Function (AF0 to AF15) for pin 8-15
Data	Input data register	GPIO x _IDR	RO	0x10	Reads the current logic level on each pin
	Output data register	GPIO x _ODR	RW	0x14	Writes the logic level to output pin
	Bit set/reset register	GPIO x _BSRR	WO	0x18	Allows atomic setting/resetting of output bits
	Bit reset register	GPIO x _BRR	WO	0x28	Allows atomic resetting of output bits

SYSTEM LEVEL OVERVIEW





CONCLUSION

- What is GPIO?
- GPIO Organization
- GPIO Registers